



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO2_AT1

Case Study Title: Flood Early Warning System

(SDG 13: Climate Action)

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1. Scenario Understanding and Problem Analysis

Business Domain

Disaster Management / Climate Monitoring / Emergency Response System

Problem Statement

Heavy rainfall and overflowing rivers cause severe flooding in many districts every year, leading to loss of human lives, damage to infrastructure, destruction of agricultural land, and disruption of transportation. Existing flood warning systems mainly depend on television, radio announcements, and manual communication, which often fail to reach all residents, especially those living in remote areas. Due to delayed warnings, people are unable to evacuate safely before floods occur. Disaster management authorities also face difficulties in monitoring multiple flood-prone locations, coordinating rescue teams, and managing emergency resources efficiently. Therefore, there is a need for an intelligent Flood Early Warning System that provides real-time flood monitoring, AI-based flood prediction, instant warning alerts, evacuation guidance, and emergency resource coordination to improve disaster preparedness and public safety.

Stakeholders

- Disaster Management Authority
- System Administrator
- Weather Monitoring Department
- Rescue Teams
- Local Government Authorities
- Citizens
- Emergency Medical Services
- Police Department
- Fire and Rescue Department
- Meteorological Department

Existing Challenges

- Delayed flood warning announcements
- Lack of real-time flood monitoring
- Dependence on television and radio communication
- Poor coordination between emergency response teams
- Difficulty in tracking affected areas
- Inefficient evacuation planning
- Lack of centralized disaster information
- Limited accessibility for remote villages
- Delayed emergency resource allocation

- Increased loss of life and property

System Objectives

- Monitor rainfall and river water levels continuously.
- Predict flood risks using Artificial Intelligence.
- Generate early warning alerts for citizens.
- Provide real-time evacuation guidance.
- Coordinate emergency rescue operations.
- Improve communication between disaster management authorities.
- Track flood-prone locations using GPS.
- Generate analytical reports for future disaster planning.
- Reduce loss of life and property.
- Improve disaster preparedness and emergency response.

2. Functional Requirements

FR No	Functional Requirement
FR1	User login and authentication.
FR2	Register weather monitoring stations.
FR3	Monitor rainfall data in real time.
FR4	Monitor river water levels.
FR5	Predict flood risks using AI algorithms.
FR6	Generate flood warning alerts automatically.
FR7	Track flood-affected locations using GPS.
FR8	Provide evacuation route guidance.
FR9	Manage emergency shelters.
FR10	Coordinate rescue teams and emergency resources.
FR11	Generate daily, weekly, and monthly reports.
FR12	Maintain historical flood records.
FR13	Send notifications to citizens and authorities.
FR14	Dashboard for monitoring overall flood conditions.

3. Non-Functional Requirements

Performance

- The system should respond within **3 seconds**.
- Support monitoring of thousands of weather sensors simultaneously.
- Process real-time weather data without delay.

Security

- Secure login with authentication.
- Role-based access control.
- Encrypted data transmission.

Reliability

- **99.9% system availability.**
- Automatic backup of weather and flood data.
- Continuous monitoring during emergency situations.

Usability

- Simple and user-friendly interface.
- Easy navigation for disaster management authorities and rescue teams.
- Mobile-friendly access for citizens.

Scalability

- Easily expandable to multiple districts and states.
- Support additional sensors and monitoring stations.
- Handle increasing numbers of users during emergencies.

Maintainability

- Modular software architecture.
- Easy software updates and bug fixing.
- Simple maintenance of weather monitoring modules.

Availability

- **24×7 system availability.**
- Cloud-based access from anywhere.
- High availability during disaster situations.

4. Actors and Use Cases

Primary Actors

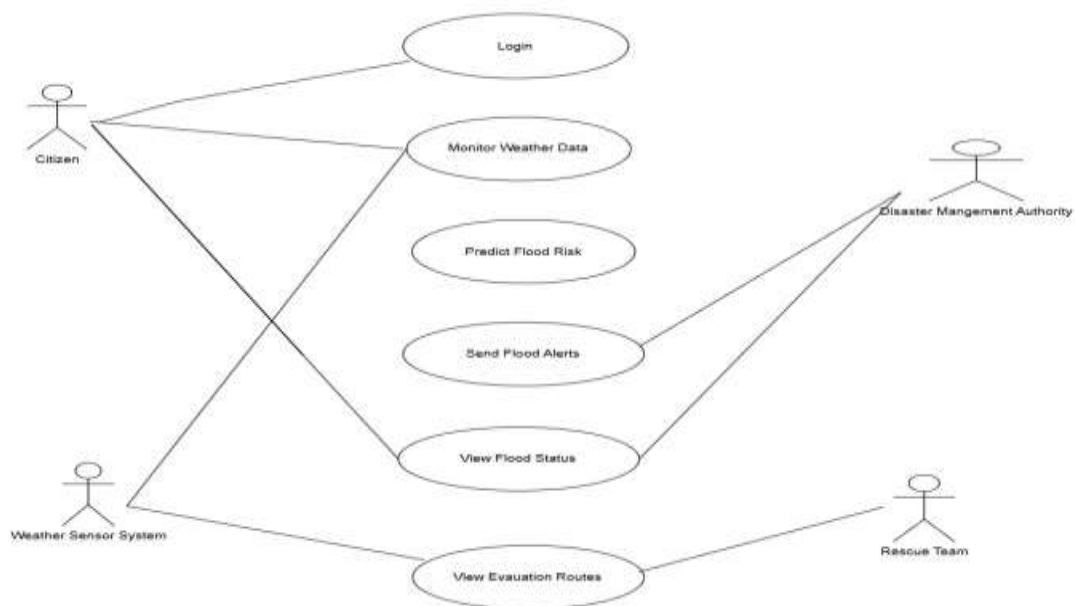
- Disaster Management Administrator
- Rescue Team

- Citizen

Secondary Actors

- Weather Sensor
- River Level Sensor
- GPS Service
- Database Server
- Meteorological Department

5. Use Case Diagram (Text Representation)



6. Software Requirement Specification (SRS)

6.1 Introduction

The **Flood Early Warning System** is a software solution developed to reduce the impact of floods by providing timely warnings, evacuation guidance, and emergency coordination. The system integrates weather sensors, river water level monitoring devices, satellite information, GPS technology, cloud computing, and Artificial Intelligence to continuously monitor flood-prone regions. By analyzing real-time environmental data, the system predicts flood risks and immediately alerts disaster management authorities and citizens. This enables faster decision-making, minimizes damage to property, reduces casualties, and improves disaster preparedness. The proposed solution also supports **SDG 13: Climate Action** by strengthening resilience against climate-related disasters.

6.2 Problem Description

Floods occur frequently during heavy rainfall and have become one of the most destructive natural disasters in many regions. Existing warning mechanisms mainly depend on television, radio announcements, and manual communication, which often fail to reach all residents quickly, especially those living in remote villages or travelling. Delayed information results in loss of lives, damage to infrastructure, and ineffective rescue operations. Disaster management authorities also face challenges in monitoring multiple flood-prone locations simultaneously. Therefore, an intelligent software system capable of predicting flood risks, issuing real-time alerts, and coordinating emergency resources is required to improve disaster response and public safety.

6.3 Scope of the System

The system will:

- Monitor rainfall and river water levels.
- Predict flood risks using AI models.
- Generate early warning alerts.
- Track affected locations.
- Guide citizens to safe evacuation centers.
- Coordinate emergency rescue operations.
- Generate reports and historical analysis.

6.4 Functional Requirements

- User authentication
- Weather sensor monitoring
- Rainfall monitoring
- River water level monitoring
- AI-based flood prediction
- Flood risk analysis
- Alert notification services
- Evacuation route guidance
- Emergency resource coordination
- Disaster management dashboard
- Report generation
- Historical flood data management
- GPS location tracking

- User role management

6.5 Non-Functional Requirements

- High performance
- Secure communication
- Reliable operation
- User-friendly interface
- Cloud scalability
- High availability
- Data integrity
- Easy maintainability

6.6 Use Case Descriptions

Use Case 1 – Monitor Flood Conditions

Actor: Disaster Management Administrator

Precondition: The administrator has successfully logged into the system.

Flow:

- Administrator opens the monitoring dashboard.
- System collects rainfall and river level data.
- AI analyzes weather information.
- Administrator reviews flood risk status.
- System highlights high-risk locations.

Postcondition:

Flood monitoring information is updated and displayed successfully.

Use Case 2 – Generate Flood Warning Alert

Actor: Disaster Management Administrator

Precondition: The system detects high flood risk.

Flow:

- AI predicts possible flooding.
- System verifies weather and sensor data.
- Warning alert is generated automatically.
- Notifications are sent to citizens and emergency authorities.

- Evacuation guidance is displayed.

Postcondition:

Flood warning alerts are successfully delivered.

Use Case 3 – Generate Food Safety Alert

Actor: Rescue Team

Precondition: Flood warning has been issued.

Flow:

- Rescue team logs into the system.
- System displays affected locations.
- Team views evacuation routes.
- Emergency resources are assigned.
- Rescue status is updated after completion.

Postcondition:

Emergency rescue activities are successfully coordinated.

6.7 Data Requirements

Inputs

- Rainfall data
- River water level
- Weather sensor readings
- GPS location
- Temperature
- Humidity
- User credentials
- Citizen reports
- Emergency resource details

Outputs

- Flood warning alerts
- Flood risk reports
- Evacuation routes
- Dashboard statistics
- Rescue operation status
- Historical flood reports
- Notification status

Database Entities

- Users
- Weather Sensors
- River Sensors
- Flood Alerts
- Citizens
- Rescue Teams
- Shelters
- Emergency Resources
- Reports
- Notifications

6.8 External Interface Requirements

User Interface

- Web dashboard for disaster management authorities
- Mobile application for citizens
- Rescue team portal
- Public emergency information portal

Hardware Interface

- Weather sensors
- River water level sensors
- GPS devices
- Cloud servers

Software Interface

- Cloud database
- AI prediction module
- Weather API
- Notification service
- Authentication module

Communication Interface

- Internet
- Wi-Fi
- 4G / 5G Network
- Email Service
- SMS Notification Service

6.9 System Features

- Real-time weather monitoring

- AI-based flood prediction
- Flood risk analysis
- Early warning alert system
- GPS-based evacuation guidance
- Rescue team coordination
- Disaster management dashboard
- Notification system
- Report generation
- Secure user authentication

7. Assumptions and Constraints

Assumptions

- Weather sensors provide accurate data.
- River level sensors operate continuously.
- Internet connectivity is available.
- GPS devices function correctly.
- Disaster management staff are trained to use the system.
- Cloud infrastructure is continuously available.

Constraints

Technical

- Depends on weather sensor accuracy.
- Requires stable internet connectivity.
- AI prediction depends on historical weather data.

Operational

- Weather sensors must be maintained regularly.
- Emergency teams must respond immediately after alerts.
- Disaster management authorities should monitor the dashboard continuously.

Economic

- Initial cost of weather sensors and GPS installation.
- Cloud hosting and maintenance costs.
- System maintenance expenses.

Legal

- Compliance with disaster management regulations.
- Protection of citizen information and location data.
- Compliance with government emergency response policies.

Time

The project should be completed within the planned development schedule.

8. Requirement Validation

Completeness

- Functional and non-functional requirements are identified.
- All stakeholders are considered.
- Major use cases are included.

Consistency

- Requirements are clear and non-conflicting.
- Functional requirements support system objectives.

Ambiguity Check

- Requirements are specific and measurable.
- User roles and responsibilities are clearly defined.

Missing Requirements Check

- Weather monitoring included.
- AI flood prediction included.
- Alert notification system included.
- Evacuation guidance included.
- Emergency resource coordination included.
- Reporting module included.

Suggested Improvements

- Integrate satellite image analysis for improved prediction.
- Add multilingual support.
- Enable offline mode for rescue teams.
- Include drone-based flood monitoring.
- Integrate social media emergency notifications.

Conclusion:

The proposed **Flood Early Warning System** provides an intelligent software solution for monitoring weather conditions, predicting flood risks, generating real-time warning alerts, and coordinating emergency response activities. By integrating Artificial Intelligence, IoT-enabled weather sensors, GPS technology, cloud computing, and mobile notifications, the system helps disaster management authorities and citizens respond quickly to flood

situations. The proposed solution reduces loss of life and property, improves disaster preparedness, enhances emergency coordination, and supports **SDG 13: Climate Action** by promoting climate resilience and sustainable disaster management.